

Yatiwelle Koralalage Samuditha Sandaru Yatiwella

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Academic Qualifications

- ❖ **University of South Carolina | Columbia, SC**
Master of Science in Chemistry - Summer 2023 | **GPA 3.864/4.00**
- ❖ **University of Colombo | Sri Lanka** (<http://www.cmb.ac.lk>)
Bachelor of Science (Honors) in Computational Chemistry | 2014-2017
GPA 3.04/4.00

Research Experience

- ❖ **University of South Carolina - Research Project – 2019-2023**
“Role of deformable mirror in multidimensional population-period transient spectroscopy”
 - Phase modulation using deformable mirror
 - Designing computer- based optical system using ray tracing program and budling up the optical setup**Supervisor:** Prof. Mark Berg, Professor,
Department of Chemistry and Biochemistry, University of South Carolina

“Machine learning classification to predict the dimensionality of hybrid inorganic halide perovskites”
 - Modeling and analyzing different machine learning models to predict different properties on different organic chemical systems using smiles strings
 - Data cleaning and processing and handling
 - Quantum chemistry DFT calculations**Supervisor:** Prof. Christopher Sutton, Assistant Professor,
Department of Chemistry and Biochemistry, University of South Carolina
- ❖ **University of Colombo - Research Project – 2017**
“Theoretical investigation of Intercalation ability of small molecules with Graphene”
 - Built optimized Graphene models and Intercalant compounds to use in intercalation process with different DFT method to investigate stability
 - Analysis of Intercalant molecules through there electron donor acceptability, structural and electronic property
 - Calculation of stabilization energy with interlayer distance and density of states for each intercalated compound**Supervisors:** Prof. Nalin de Silva, Senior Lecturer,
Department of Chemistry, University of Colombo
Prof. Rohini De Silva, Senior Lecturer,
Department of Chemistry, University of Colombo
- ❖ **SLINTEC – Product Development for Ceylon Graphene Technologies – 2018-2019**
 - Optimization of product development process for graphene production based on lab trails followed by lab analysis in collaboration with SLINTEC engineering and science team
 - Optimization of Graphene Oxide (GO) and reduced Graphene Oxide (rGO) as commercial product
 - Analysis of vein Graphite, GO, rGO
 - Mass scale production of GO and rGO
 - Check the availability of rGO as an anti corrosive agent

- Working Experience**
- ❖ **Teaching assistant | IChem | Sri Lanka 2018**
 - Coordinating and document handling to acquire the accreditation for degree offering status
 - ❖ **Research Intern | SRI LANKA INSTITUTE OF NANOTECHNOLOGY (SLINTEC) 2018-2019**
 - Development and optimization of Graphene oxide and reduced Graphene oxide for Ceylon Graphene Technologies (CGTL)
 - Waste Management of Acids
 - Recovery of ethanol and Ether as low-grade solvents
 - Conducting research activities between SLINTEC and clients
 - ❖ **Graduate Teaching Assistant | University of South Carolina | USA 2019-2023**
 - Conducted laboratory sessions for chemistry freshman and honors students
 - Prepared laboratory materials and ensured proper setup for experiments
 - Supervised students during lab sessions, maintaining safety and supporting learning
 - Graded assignments, lab reports, and proctoring exams
 - Provided supplemental academic assistance to students outside of class
 - Mentoring ACS project seed summer program students
 - ❖ **Instructor | University of South Carolina | USA 2023-2025(present)**
 - Taught more than 20 lab sections per each academic year, for non-science majors
 - Setup labs and supervised students during lab sessions, maintaining safety standards
 - Organized and led TA meetings to explain experiments, procedures, and grading, providing instructional guidance for teaching assistants.
 - Managed chemical and instrument inventories for laboratory experiments by tracking usage, ensuring proper storage and calibration, maintaining safety compliance, and coordinating timely procurement of supplies
 - Collaborated with faculty and TAs to troubleshoot faulty instruments, standardize lab instruction, and improve consistency across multiple course sections
- Technical Skills**
- ❖ **Computational Chemistry Related:**
 - Monte Carlo Simulations
 - Molecular modeling and molecular dynamic simulations
 - Computer languages (Fortran, Matlab, Python)
 - Software packages (Gaussian 98/03/09, HyperChem, RasMol, Hex, Winmopac, ChemDraw, Rdkit, Scikit learn, Mercury, Qchem, Psi4, Lammmps, VASP, Ray tracing, Overleaf)
 - Working experience in Linux, Windows and Mac operating systems
 - Mobile application development
 - ❖ **Chemistry Related:**
 - Chromatographic techniques (Thin Layer Chromatography, Paper Chromatography, GC/MS, HPLC, Column Chromatography)
 - Spectroscopic techniques (Atomic Absorption Spectroscopy, UV/Visible spectroscopy/Colorimetry, NMR, FT-IR, Raman Spectroscopy)
 - Other Instrumentation
XPS, XRD, TGA, XRF, SEM, IC-PMS, Particle size analyzer
 - Qualitative & quantitative analysis of inorganic compounds (Cation/Anion Analysis)

	▪ Qualitative & quantitative analysis of organic compounds (Functional Group Analysis) ▪ Complexometric titrations ▪ Potentiometric / Conductivity and PH Measurements ▪ Recrystallization / Distillation / Solvent Extraction Techniques	
Presentations	▪ Conducted an oral presentation on “Photo Induced Charge Transfer” for the partial fulfillment of CH4002 – Seminars and Essay ▪ “Role of deformable mirror in multidimensional population-period transient spectroscopy”- Seminar 2020 ▪ Poster presentation on the topic “Classification of HOIPS (Hybrid Organic Inorganic Perovskite Structures) using Machine Learning ” at SETCA 2023	
Selected Extra-Curricular Activities	▪ Community Service Director Rotaract club Faculty of Science University of Colombo ▪ Member First XV Rugby Team University of Colombo ▪ Committee Member of Chemical Society University of Colombo, Sri Lanka in 2016/2017 ▪ CO-Chief organizer of project SAHAS (Annual charity project organized by Chemical Society, University of Colombo, Sri Lanka in 2016 ▪ University colors – 2016,2017	
Trainings attended	▪ Leadership Training Conducted by University Grants Commission Sri Lanka ▪ XPS training Conducted by Product Specialist-XPS, Thermo Fisher Scientific India Pvt. Ltd ▪ ACS user awareness conducted by ACS representative ▪ Laser training conducted by University of South Carolina ▪ Chemical waste handling conducted by University of South Carolina	
Referees	▪ Prof. Mark A Berg, Professor Spectroscopy and Chemical Physics, Department of Chemistry and Biochemistry, University of South Carolina, USA Email: berg@sc.edu Mobile: (803)777-1514	▪ Prof. Amelia Taylor Perry, Senior Instructor, Department of Chemistry and Biochemistry, University of South Carolina, USA Email: taylora4@mailbox.sc.edu Mobile: (803)-777-1540
	▪ Dr. Leslie Lovelace, Associate chair, Department of Chemistry and Biochemistry, University of South Carolina, USA Email: lovelacl@mailbox.sc.edu Mobile: (803)-777-0458	